# CSS – Layout & Responsive Design

Web Design – Lecture 5

(C1WD1A/NWD011//VT2026)

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# Recap

- **Last time:**
  - CSS controls presentation
  - Selectors: element, class, ID
  - Specificity and the cascade
  - The box model: content, padding, border, margin

- **Today:** Positioning and layout

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# Agenda

1. Modern CSS Layout
2. Responsive Design
3. CSS variables
4. Live Coding

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# Part 1

## Modern CSS Layout

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# The Layout Challenge

- How do we arrange elements on a page?
  - Side-by-side columns
  - Centred content
  - Navigation bars
  - Card grids

- CSS has evolved to make this easier.

# A Brief History of CSS Layout

- **Tables** (1990s) — Never meant for layout
- **Floats** (2000s) — Hacky, fragile
- **Flexbox** (2012) — One-dimensional layout
- **Grid** (2017) — Two-dimensional layout

- Today we use Flexbox and Grid.

# Display Property

- Every element has a display type:

```
/* Takes full width, stacks vertically */
display: block;

/* Flows with text, no width/height */
display: inline;

/* Inline but respects width/height */
display: inline-block;

/* Hides the element */
display: none;
```

# Block *vs.* Inline

- **Block elements:** `<div>`, `<p>`, `<h1>`, `<section>`
  - Take full width
  - Stack vertically

- **Inline elements:** `<span>`, `<a>`, `<strong>`
  - Flow with text
  - Sit side by side

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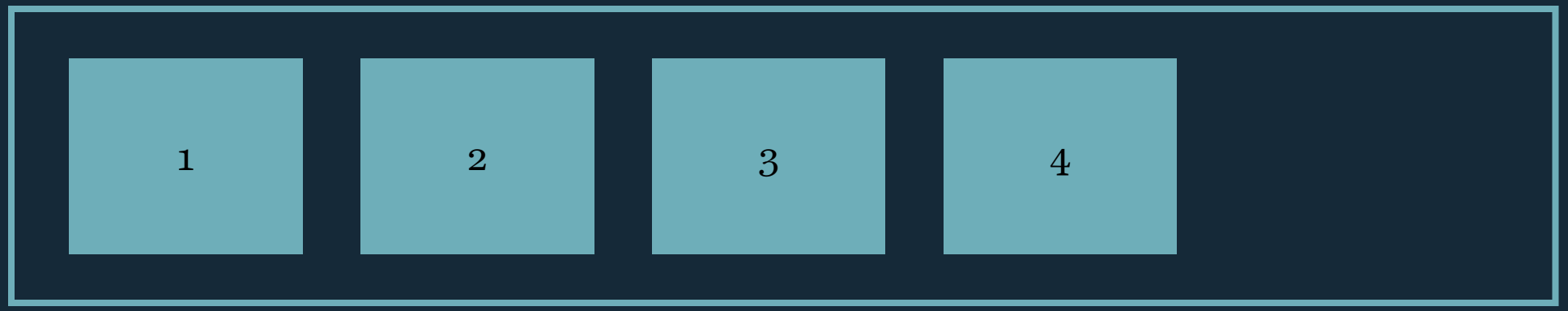
# Flexbox: The Basics

```
.container {
  display: flex;
}
```

- Children of container are ***flex items*** & line up in a row:

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# Flex Direction

```
.container {
  display: flex;
  flex-direction: row;      /* default: horizontal */
}

.container {
  display: flex;
  flex-direction: column;   /* vertical */
}
```

# Justify Content

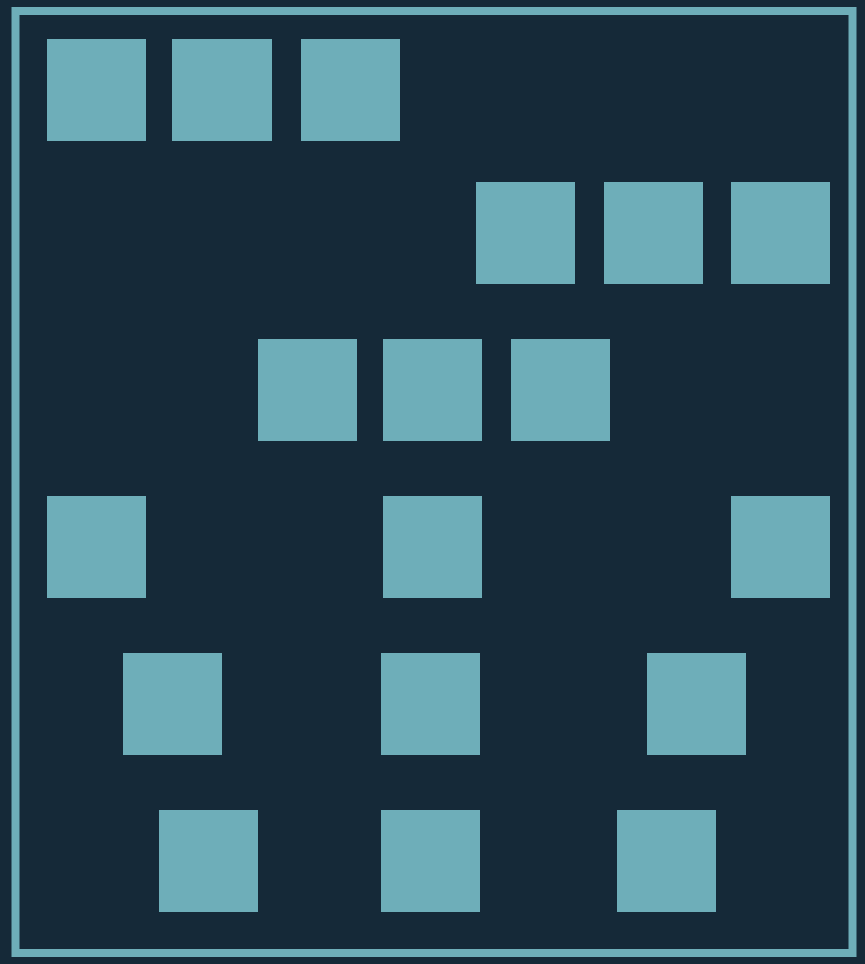
Distributes space ***along the main axis***:

```
.container {
  display: flex;
  justify-content: flex-start;   /* default */
  justify-content: flex-end;
  justify-content: center;
  justify-content: space-between;
  justify-content: space-around;
  justify-content: space-evenly;
}
```

# Justify Content

## *Visual*

- flex-start:
- flex-end:
- center:
- space-between:
- space-around:
- space-evenly:



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# Align items

- Distributes space ***along the cross axis***:

```
.container {
  display: flex;
  align-items: stretch;      /* default */
  align-items: flex-start;
  align-items: flex-end;
  align-items: center;
  align-items: baseline;
}
```

# Gap

- The space between flex items:

```
.container {
  display: flex;
  gap: 20px;
}
```

- (Cleaner than adding margins to children.)

# Centring with flexbox

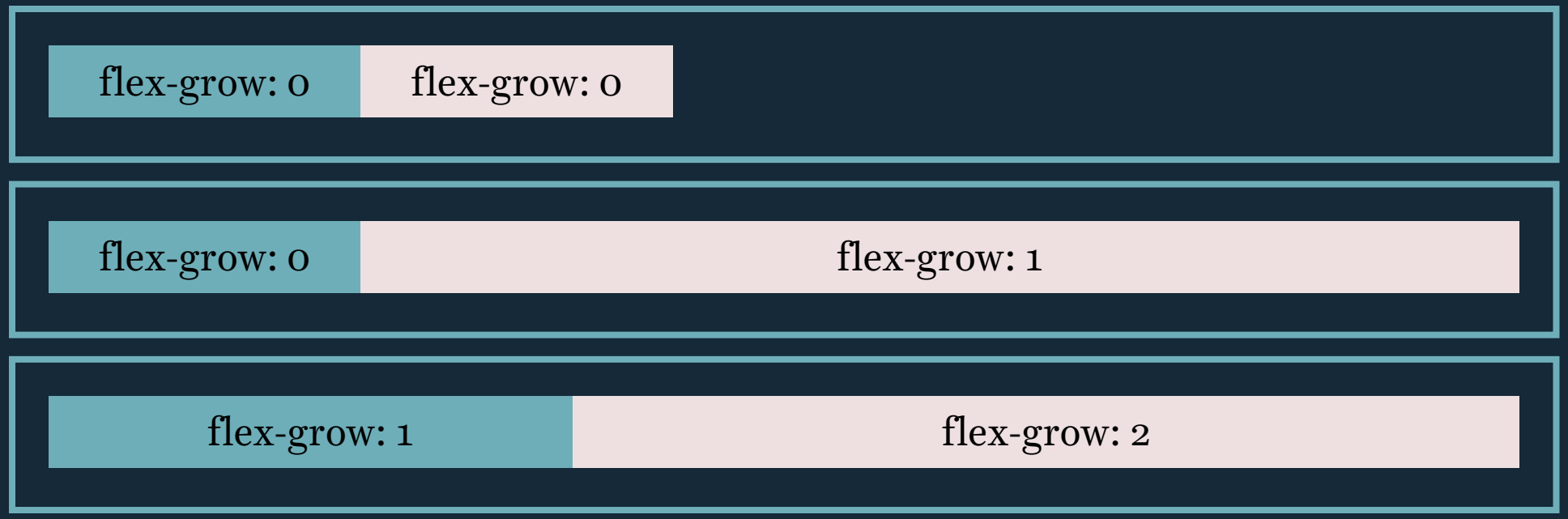
- The easiest way to centre anything:

```
.container {
  display: flex;
  justify-content: center;  /* horizontal */
  align-items: center;      /* vertical */
  height: 100vh;
}
```

# Flex items

## *growing items*

```
.item {
  flex-grow: 1;   /* Take available space */
}
```

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# Common pattern

## *Navigation bar*

```
nav {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 1rem;
}
```

---

```
<nav>
  <a href="/" class="logo">Logo</a>
  <ul class="nav-links">...</ul>
</nav>
```

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# Common pattern

## *Card Layout*

```
.card-container {
  display: flex;
  gap: 20px;
  flex-wrap: wrap;
}

.card {
  flex: 1 1 300px;  /* grow, shrink, basis */
}
```

- Cards wrap to new rows when space runs out.

# CSS Grid
## *Introduction*

- **Two-dimensional layout** – rows *and* columns

```
.container {
  display: grid;
  grid-template-columns: 1fr 1fr 1fr;
  gap: 20px;
}
```

- (Creates a 3-column grid with equal widths.)

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# Flexbox vs. Grid

- **Flexbox**
  - One dimension (row OR column)
  - Content-driven sizing
  - Good for: navigation, centering, distributing items

- **Grid**
  - Two dimensions (rows AND columns)
  - Layout-driven sizing
  - Good for: page layouts, card grids, complex arrangements

- Use both — they complement each other.

# Part 2

## Responsive Design

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# Why Responsive design?

- **Users browse on everything:**
  - Phones (320px – 480px)
  - Tablets (768px – 1024px)
  - Laptops (1024px – 1440px)
  - Desktops (1440px+)
  - TVs, etc, etc…

- One website should work on all of them.

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# Mobile traffic

- More than 50% of web traffic is mobile
- If your site doesn’t work on phones, your losing more than half your audience

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# The viewport meta tag

- Essential for mobile:

```
<meta name="viewport" content="width=device-width,
 initial-scale=1.0">
```

- Without this, small-screen browsers zoom out to show the ”desktop” version.
- Always include in your `<head>`.

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# Mobile-first vs. Desktop-first

- **Desktop-first**: design for big screens, adapt-down:

```
.container { width: 1200px; }
@media (max-width: 768px) {
  .container { width: 100%; }
}
```

- **Mobile-first**: design for small screens, adapt up:

```
.container { width: 100%; }
@media (min-width: 769px) {
  .container { width: 1200px; }
}
```

- Generally, **mobile-first recommended.**

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# Media queries

- Apply CSS only when conditions are met:

```
/* Base styles (mobile) */
.container {
  padding: 10px;
}

/* Larger screens */
@media (min-width: 768px) {
  .container {
    padding: 20px;
  }
}
```

# Common breakpoints

```
/* Mobile-first approach */
/* Small phones: default styles */

/* Large phones */
@media (min-width: 480px) { }

/* Tablets */
@media (min-width: 768px) { }

/* Laptops */
@media (min-width: 1024px) { }

/* Desktops */
@media (min-width: 1200px) { }
```

# Responsive typography

```
/* Base size for mobile */
body {
  font-size: 16px;
}

/* Larger text on bigger screens */
@media (min-width: 768px) {
  body {
    font-size: 18px;
  }
}
```

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# Responsive Layout Example

```
/* Mobile: single column */
.grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 20px;
}
/* Tablet: two columns */
@media (min-width: 768px) {
  .grid {
    grid-template-columns: 1fr 1fr;
  }
}
/* Desktop: three columns */
@media (min-width: 1024px) {
  .grid {
    grid-template-columns: 1fr 1fr 1fr;
  }
}
```

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# Flexible images

- Images should scale with their container:

```
img {
  max-width: 100%;
  height: auto;
}
```

  - `max-width: 100%` – image never exceeds container width
  - `height: auto` – maintain aspect ratio

# Relative units

- Fixed units:
  - `px` – pixels

- Relative units:
  - `%` – percentage of parent
  - `em` – relative to parent font-size
  - `rem` – relative to root (<html>) font-size
  - `vw` – percentage of viewport width
  - `vh` – percentage of viewport height

# Using relative units

```
.container {
  width: 90%;             /* 90% of parent */
  max-width: 1200px;      /* but never more than 1200px */
  margin: 0 auto;         /* centered */
}

h1 {
  font-size: 2rem;        /* 2× root font size */
}

.hero {
  height: 100vh;          /* full viewport height */
}
```

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# Testing responsive designs

- **Browser DevTools (F12):**
  - Toggle device toolbar
  - Test different screen sizes
  - Simulate devices

- **Test on real devices too!**

# Accessibility & CSS

- **Colour contrast:**
  - Text must be readable
  - WCAG: 4.5:1 ratio for normal text
  - Use contrast checker tools

- **Focus states:**

```
a:focus, button:focus {
  outline: 2px solid blue;
}
```

- Never remove focus outlines without replacement.

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# Hiding content accessibly

```
/* Hidden from everyone */
.hidden {
  display: none;
}

/* Hidden visually, available to screen readers */
.sr-only {
  position: absolute;
  width: 1px;
  height: 1px;
  clip: rect(0, 0, 0, 0);
  overflow: hidden;
}
```

# Part 3

## CSS Variables

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# CSS Variables
## *Custom Properties*

- Define reusable values once, use them everywhere.

```
:root {
  --primary-color: #3498db;
  --spacing: 1rem;
}

header {
  background-color: var(--primary-color);
  padding: var(--spacing);
}

button {
  background-color: var(--primary-color);
  margin: var(--spacing);
}
```

- Change `--primary-color once` → updates everywhere.

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# Why use CSS Variables?

- **Consistency**
  - Same values used throughout your stylesheet
- **Maintainability**
  - Change a colour or spacing in one place
- **Theming**
  - Easy dark mode, brand variations
- **Readability**
  - `var(--primary-color)` is clearer than `#3498db`
-

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# Syntax

- **Define variables** (usually in `:root` for global scope):

```
:root {
  --name: value;
}
```

- Use variables:

```
property: var(--name);
```

- With fallback:

```
property: var(--name, fallback-value);
```

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# Common Use Cases

```
:root {
  /* Colours */
  --color-primary: #3498db;
  --color-text: #333;
  --color-background: #fff;

  /* Typography */
  --font-body: 'Open Sans', sans-serif;
  --font-heading: 'Roboto', sans-serif;

  /* Spacing */
  --spacing-sm: 0.5rem;
  --spacing-lg: 2rem;

  /* Other */
  --border-radius: 4px;
  --shadow: 0 2px 4px rgba(0,0,0,0.1);
}
```

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# Dark Mode Example

```
:root {
  --color-text: #333;
  --color-background: #fff;
}

@media (prefers-color-scheme: dark) {
  :root {
    --color-text: #f0f0f0;
    --color-background: #1a1a1a;
  }
}

body {
  color: var(--color-text);
  background-color: var(--color-background);
}
```

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# Part 4

Live coding

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# Live Coding

## *Adding layout to an HTML page*

- **Let's Add Layout**
  - We'll make our course page responsive using Flexbox and media queries.

- **We'll cover:**
  - Flexbox for header (title + nav side by side)
  - Responsive navigation (stacks on mobile)
  - Two-column layout for wider screens
  - A media query breakpoint

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# Part 5

## Conclusion

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# Key takeaways

- Flexbox for one-dimensional layout
- Grid for two-dimensional layout
- Mobile-first: design for small screens, enhance for larger
- Media queries adapt styles to screen size
- Use relative units and flexible images
- Always include the viewport meta tag

# Looking ahead

- **Next**: JavaScript and Interactivity
  - Adding behaviour to your pages
  - The DOM
  - Events and event listeners

# Questions?

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